Dockerized FastAPI Application Setup

Step1: Fork and Clone the given Repository

CMD: gitclone https://github.com/RohitPatil18/docker-fastapi-test

Step2: Create a Dockerfile and docker-compose.yml file

CMD: vi Dockerfile

```
# Using the official Python Image
FROM python:3.9
# Set the working directory in the container
WORKDIR /app
# Copy the requirements.txt file and install the Python dependencies
COPY requirements.txt ./app
RUN pip install --no-cache-dir -r requirements.txt
# Copy the application files to the container
COPY . /app
# Expose the necessary port
EXPOSE 8000
# Define the command to run the application
CMD ["uvicorn", "main:app", "--host", "0.0.0.0", "--port", "8000"]
```

CMD: vi docker-compose.yml

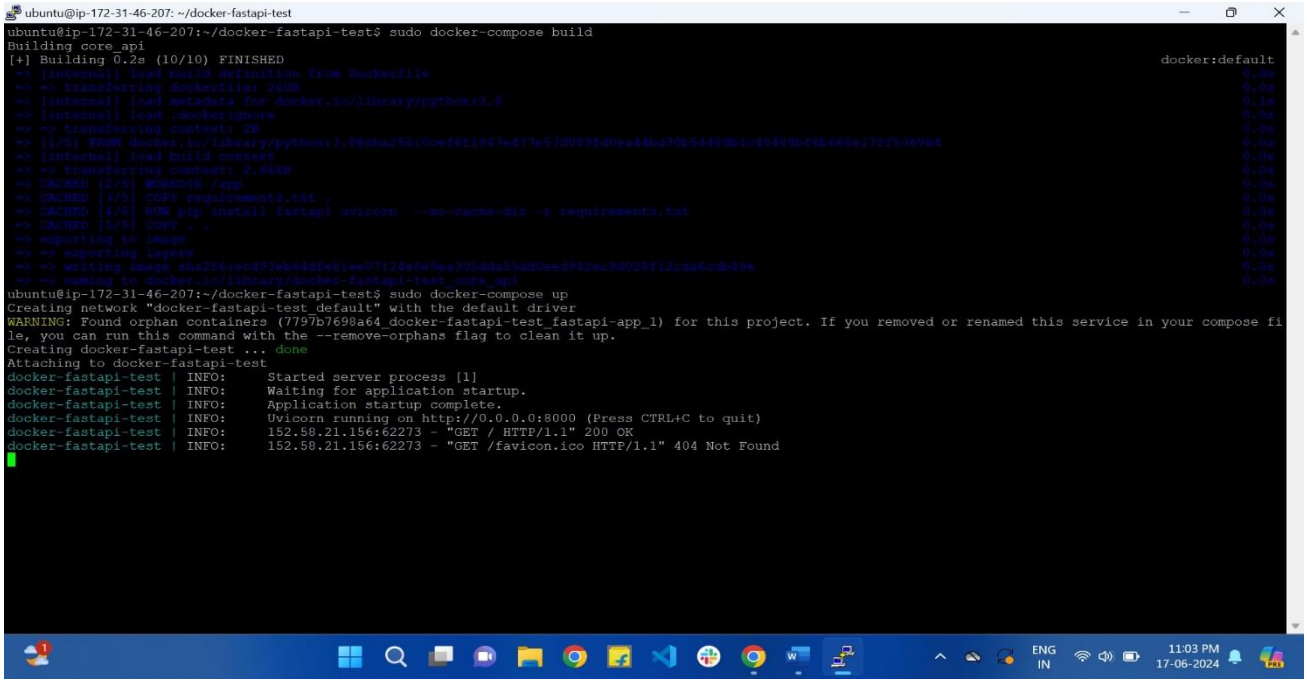
```
version: "3.9"
services:
  core_api:
    build: .
    container_name: "docker-fastapi-test"
    ports:
      - "8000:8000"
    volumes:
```

- ./:/app

Step3: Create a Container by using above Docker Files

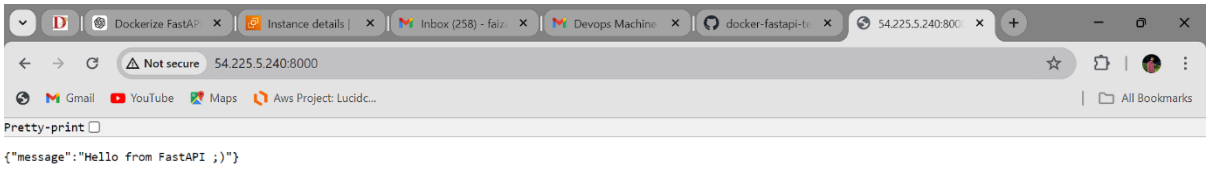
CMD: sudo docker-compose build

sudo docker-compose up

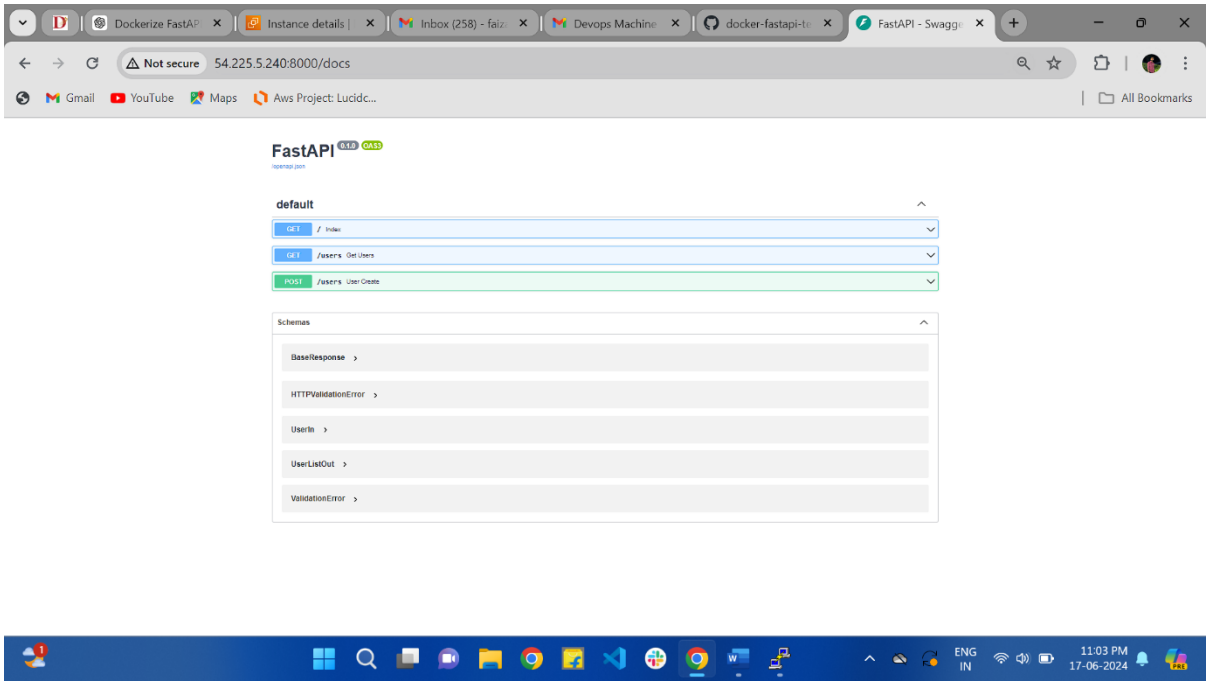


Step4: Check the Containerized App Accessible on Web

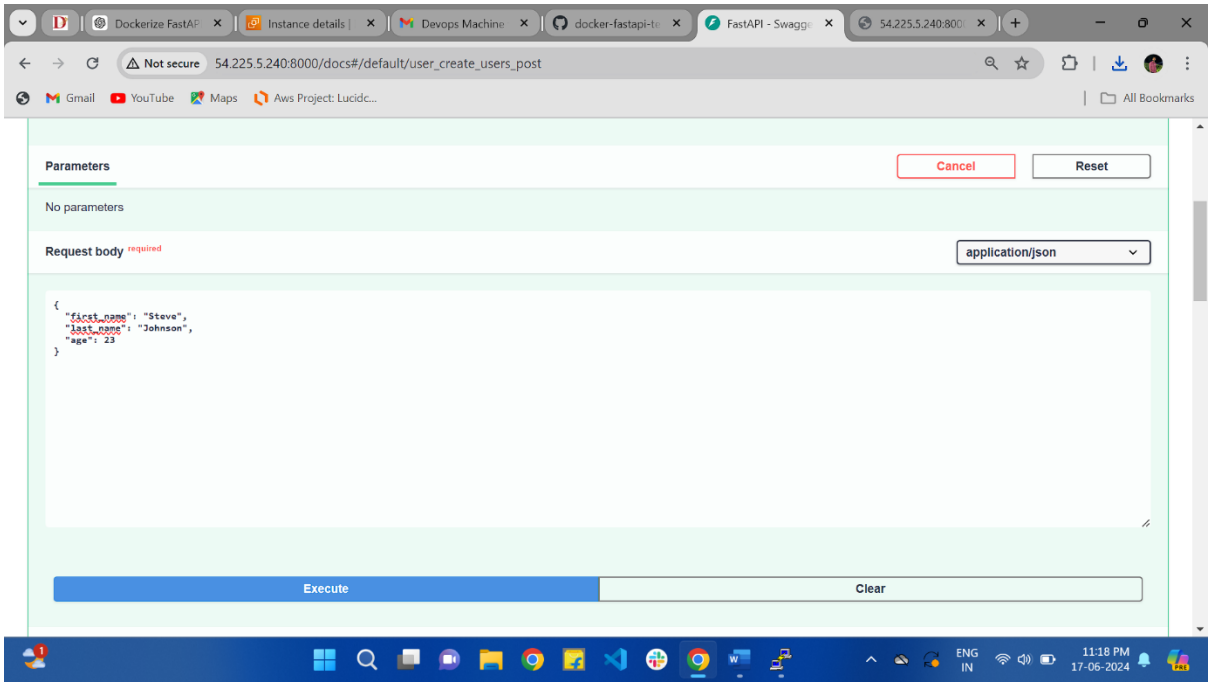
IP 54.225.5.240:8000



IP 54.225.5.240:8000/docs



Accepts and stores user data in JSON file

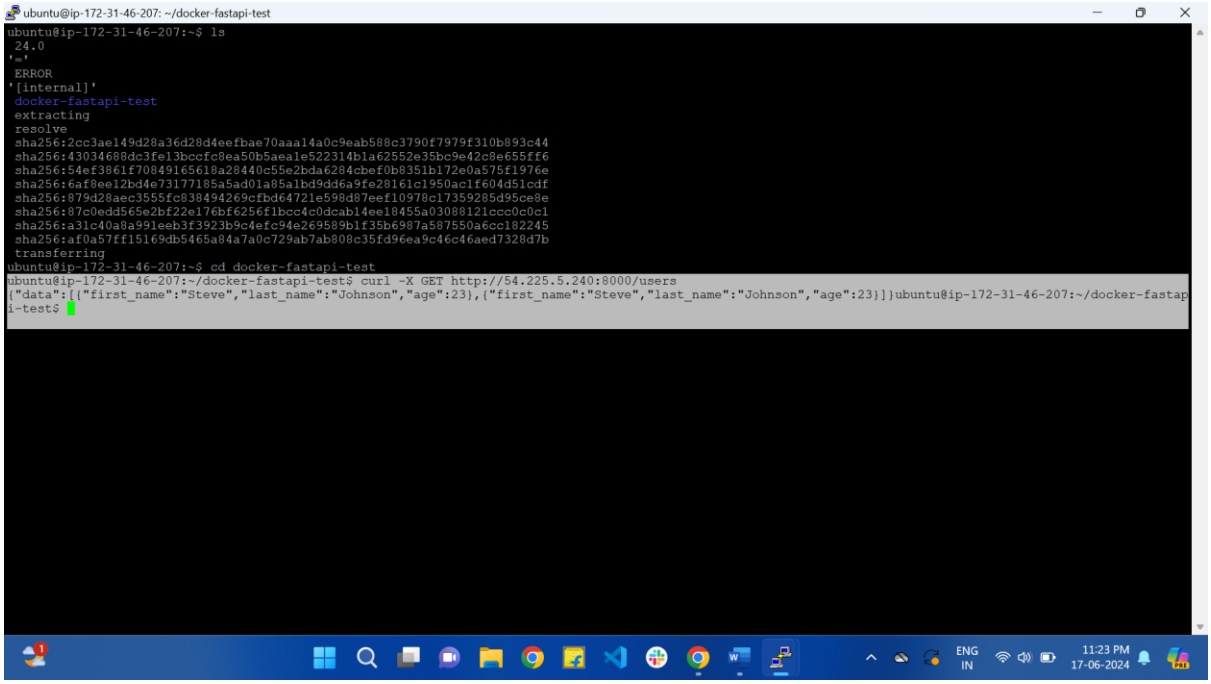


Step5: Stop the Container and check Data Persistence

CMD: docker-compose down

Step6: Check the data/users.json File: Ensure that the file contains the data previously added

Curl -X GET http:// 54.225.5.240:8000/users



***Step7*:** Commit and push changes to your forked repo

CMD: git add.

git commit -m “Fast-API app”

git push origin main.

<u>ThankYou</u>